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Underwater Object Detection Algorithm Based on an Improved YOLOv8

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Abstract: Due to the complexity and diversity of underwater environments, traditional object detection algorithms face challenges in maintaining robustness and detection accuracy when applied underwater. This paper proposes an underwater object detection algorithm based on an improved YOLOv8 model. First, the introduction of CIB building blocks into the backbone network, along with the optimization of the C2f structure and the incorporation of large-kernel depthwise convolutions, effectively enhances the model's receptive field. This improvement increases the capability of detecting multi-scale objects in complex underwater environments without adding a computational burden. Next, the incorporation of a Partial Self-Attention (PSA) module at the end of the backbone network enhances model efficiency and optimizes the utilization of computational resources while maintaining high performance. Finally, the integration of the Neck component from the Gold-YOLO model improves the neck structure of the YOLOv8 model, facilitating the fusion and distribution of information across different levels, thereby achieving more efficient information integration and interaction. Experimental results show that YOLOv8-CPG significantly outperforms the traditional YOLOv8 in underwater environments. Precision and Recall show improvements of 2.76% and 2.06%. Additionally, mAP50 and mAP50-95 metrics have increased by 1.05% and 3.55%, respectively. Our approach provides an efficient solution to the difficulties encountered in underwater object detection.

Keywords: underwater recognition; YOLOv8; object detection

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1. Introduction

Underwater object detection, a pivotal aspect of machine vision, has achieved remarkable advancements across various disciplines in recent years. This technology is particularly vital in marine science research and military applications. In marine science, underwater object detection is extensively employed for monitoring marine organisms, exploring seabed topography, and contributing to environmental conservation efforts. Within the military sector, this technology is instrumental in the navigation of underwater vehicles, as well as in the reconnaissance and surveillance of hostile targets. However, the intricate and dynamic nature of underwater environments presents considerable challenges to effective object detection.

Primarily, the phenomena of light absorption and scattering in water lead to image degradation through foggy blurring and color distortion. These optical effects obscure the edges of underwater objects and cause them to blend with complex backgrounds, thus complicating detection efforts. Moreover, additional interference factors such as water quality [1], plankton, and other particulates in the water further diminish image clarity and reduce the accuracy of target identification. Consequently, developing robust underwater object detection technologies necessitates overcoming a range of technical hurdles. Researchers must devise algorithms capable of compensating for the unique optical properties of underwater environments while integrating cutting-edge image processing

and deep learning techniques to enhance detection precision and reliability. Furthermore, it is essential to develop specialized detection equipment and systems tailored to specific applications, ensuring that the solutions are well-suited to the demands of marine science and military operations.

In summary, underwater object detection technology is indispensable for the advancement of marine science research and military capabilities. With ongoing technological advancements, we can expect this field to achieve even greater success in the future, broadening its application potential and providing robust solutions for an increasing array of practical challenges.

With the rapid advancement of deep learning technologies [2] and their widespread application across various fields, deep learning-based object detection algorithms have demonstrated excellent performance on general datasets. These algorithms have effectively addressed many traditional object detection challenges, yet there remain unresolved issues within underwater datasets. Currently, mainstream object detection algorithms are primarily categorized into two types: the "two-stage detectors", which include region proposal-based networks such as Region-based Convolutional Neural Networks (R-CNN, 2014) [3], Fast R-CNN [4], and Faster R-CNN [5] and the "single-stage detectors", exemplified by the Single Shot MultiBox Detector (SSD) [6] and the "You Only Look Once" (YOLO) series [7].

Hsiao et al. [8] developed a maximum probability bias ranking method based on sparse representation (SRC-MP), which effectively extracted features using feature surfaces and Fisherfaces, significantly improving the accuracy of real underwater observation video data. Li et al. [9] applied the Fast R-CNN model to a fish dataset, achieving promising results; however, its detection performance on underwater image datasets was lower compared to general above-water datasets. Salman et al. [10] proposed a deep learning method for fish classification in unconstrained underwater environments, significantly enhancing classification accuracy through the hierarchical feature combination of convolutional neural networks. Zhuang et al. made breakthrough progress in the SEACLEF-2017 challenge [11], where they employed advanced deep learning models, using pre-trained networks and the BN-Inception network to significantly improve the accuracy of marine organism identification. Zhao et al. [12] introduced an improved motion influence map combined with a recurrent neural network (RNN), which substantially enhanced the detection and recognition accuracy of localized abnormal behaviors in densely populated aquaculture environments. Zhou et al. [13] developed an automated evaluation approach utilizing convolutional neural networks (CNN) and machine vision for assessing fish feeding intensity. This method leveraged data augmentation techniques and CNN model training to greatly improve evaluation accuracy. Adiwinata et al. utilized Faster R-CNN for fish recognition and detection [14]. Although two-stage object detection methods offer slightly higher accuracy, they involve more complex processing, have poorer adaptability to scene changes, and exhibit slower detection speeds.

Single-stage detection methods bypass the region proposal phase, directly performing dense sampling at potential object locations to achieve detection. This approach enables a faster and more streamlined detection process, though it may come at the expense of some performance. Given its distinct speed advantage, we have chosen to employ a single-stage detector as the baseline for underwater object detection. The YOLO series, recognized as a seminal algorithm within single-stage methodologies, is known for its speed and strong real-time capabilities, making it widely adopted within the domain of object detection. Zhao et al. [15] enhanced the YOLOv3 model for vehicle detection, specifically improving its ability to detect smaller vehicles. Lu et al. [16] introduced an augmented model based on YOLOv5, achieving significant improvements in distinguishing cracks and large objects like vehicles while maintaining the original YOLOv5 model's real-time capabilities. Wang et al. [17] advanced small object detection in aerial imagery by utilizing the CoT-YOLOv8 algorithm, leading to a marked performance improvement. With the YOLO series' significant progress in visual detection, numerous researchers have introduced various enhancements to better adapt it for underwater object detection tasks. Cai

et al. [18] proposed an underwater fish detection method that integrates YOLOv3 with MobileNetv1, significantly boosting detection accuracy by optimizing feature map selection. Yu et al. [19] presented an enhanced U-YOLOv7 framework built upon the original YOLOv7 model, designed for efficient and accurate underwater biological detection. Cai et al. [20] proposed a novel NAM-YOLOV7 method for rapid detection of fish exhibiting symptoms of SVC, utilizing a NAM attention mechanism to precisely extract features and significantly enhance detection efficiency. Yi et al. [21] addressed the challenges of missed detections and low detection accuracy in underwater small object detection by developing an enhanced UWSC-YOLOv7 method for recognizing tiny underwater organisms. The network enhances detection accuracy and reduces model complexity by incorporating the SENet attention mechanism, optimizing the FPN architecture, and utilizing the EIoU loss function to better capture essential features of small objects. Qu [22] introduced a new neural network model, YOLOv8-LA, aimed at improving underwater object detection performance. This model enhances spatial feature extraction by selectively processing input channels and combines depthwise separable convolutions with varying dilation rates with FasterNet, enhancing the model's ability to capture the fine details of small objects while minimizing information loss during the upsampling process. This model maintains real-time processing capabilities while ensuring high detection accuracy. Zhu et al. [23] proposed an underwater object detection algorithm based on YOLOv8, replacing the original C2f module with Dynamic Snake Convolution (DSConv) to better handle the overlap between objects and backgrounds in underwater images. They also introduced continuity constraints in the design of convolutional kernels to enhance perception. These improvements significantly boost the detection performance of the YOLOv8 model in complex underwater environments. YOLOv8 [24] represents an advanced object detection method, showcasing immense potential in underwater detection tasks due to its rapid and accurate performance. Not only does YOLOv8 achieve notable precision improvements, but it also requires considerably fewer resources compared to transform-based structures. Its core strength lies in its ability to accurately detect and localize objects while maintaining a high detection speed, which is particularly critical for real-time underwater operations. The application of YOLOv8 in underwater object detection tasks can significantly enhance both detection efficiency and accuracy, thereby increasing overall operational success rates and effectiveness.

Although the current YOLOv8 model performs well on standard datasets, its accuracy in detecting distant small objects and underwater targets still requires improvement. Underwater target detection is a highly challenging task, as the dim and murky underwater environment, combined with reduced contrast and information density, hinders precise target localization in such scenes. The existing research presents several key challenges: first, the use of small convolutional kernels leads to suboptimal detail extraction, resulting in poor recognition performance; second, the varying distances of underwater targets cause inadequate attention to feature extraction, leading to insufficient feature fusion.

To tackle these issues, this paper introduces an improved model for underwater small object detection, built upon the YOLOv8 framework, named YOLO8-CPG, which effectively tackles the aforementioned issues. Our contributions are summarized as follows:

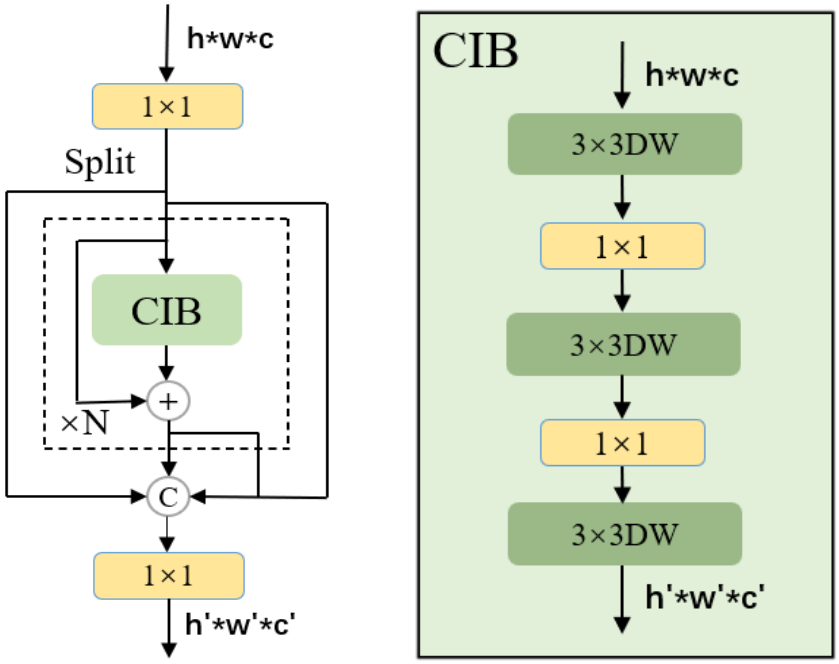
- Due to environmental interference, feature extraction and precise detection are hindered. By integrating the C2f module into the backbone network, we enhance the model's ability to more effectively recognize target contours, thereby improving detection accuracy and efficiency while maintaining feature extraction capability and accelerating model inference speed.
- We introduce a Partial Self-Attention (PSA) module into the final layer of the backbone network to strengthen focus on underwater target features, enhancing the model's capability to extract critical features, leading to greater detection accuracy and overall performance.
- To address the issue of small underwater targets, we employ the low-level collection-distribution mechanism from the Gold-YOLO architecture to refine the neck structure

of YOLOv8. This enhancement bolsters multi-scale feature fusion capabilities, thereby increasing the accuracy of target detection.

2. Methods

2.1. Compact Inverted Block

As the depth of neural networks increases, the introduction of deeper layers and larger models often leads to an accumulation of redundancy, which can degrade model efficiency and negatively impact the accuracy of target detection. To address this issue, Wang et al. [25] proposed a Compact Inverted Block (CIB). By employing a rank-based block allocation strategy, they evaluated the performance of substituting the CIB module across different stages. This approach effectively reduces redundancy without compromising accuracy, thereby achieving higher efficiency while maintaining model performance. The structure of CIB module is shown in Figure 1.



(**a**) (**b**)

Figure 1. Structure diagram of the CIB. (**a**) CIB deployment; (**b**) Internal structure of the CIB.

The Compact Inverted Block (CIB) module leverages three efficient depthwise convolutions for spatial mixing and two cost-effective pointwise convolutions for channel mixing. The use of multiple depthwise convolution layers in spatial mixing enables the extraction of richer and more complex feature representations. Each convolution layer is capable of capturing features at different scales and abstraction levels, integrating and expressing information across various layers of the network. This hierarchical feature extraction process enhances the network's generalization and representation capabilities, allowing for a more nuanced understanding and analysis of input data. Depthwise convolutions, which have fewer parameters or reduced computational requirements, significantly lower the model's computational cost, accelerating both training and inference while maintaining accuracy. Pointwise convolutions, compared to traditional convolution operations, involve fewer parameters and reduced computational overhead. The use of 1×1 convolutions allows for the reduction in the number of channels while preserving the spatial dimensions of the feature maps, thereby minimizing computational resource consumption. Additionally, these convolutions can introduce non-linear transformations through activation functions such as ReLU, enhancing the network's ability to mix channels and introduce more complex feature expressions. This improvement is crucial for bolstering the model's perceptual and

generalization capabilities and augmenting its non-linear expression potential. The CIB module can serve as an efficient fundamental building block, such as when embedded within the ELAN structure. In this study, we improved the original C2f structure in the backbone network by replacing all Bottleneck layers with CIB modules, resulting in a new CIBC2f structure. In the deeper stages of the backbone network, we employed large kernel depthwise convolutions within the CIB module. Specifically, in the final CIBC2f module at the deeper stages, the kernel size of the second 3 × 3 depthwise convolution in the CIB block was increased to 7 × 7. This use of large kernel depthwise convolutions expands the receptive field and enhances the model's capabilities, leading to better detection performance. Furthermore, we employed structural re-parameterization techniques by introducing an additional 3 × 3 depthwise convolution branch, which does not increase inference overhead but helps alleviate optimization challenges. As shown in Figure 2.

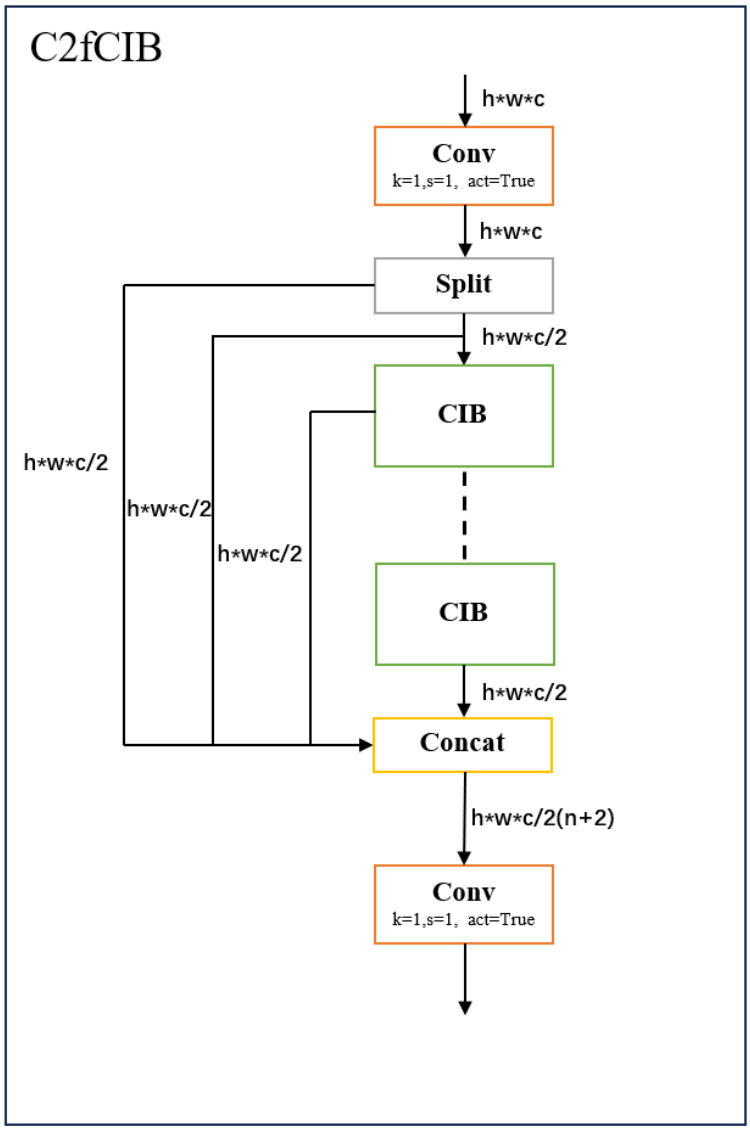


Figure 2. Structure diagram of the CIBC2f.

The input feature map is initially processed by a convolutional (Conv) layer to extract the fundamental features of the input image, resulting in an intermediate feature map. This intermediate feature map is then bifurcated into two segments: one is directly forwarded to the final Concat block, while the other undergoes further processing through multiple CIB blocks. The feature map fed into the CIB is subjected to a series of convolutions, normalization, and activation operations to refine and enhance the features, capturing more

intricate patterns and details. The processed feature map is subsequently concatenated with the feature map that was directly passed to the Concat block, enabling the model to leverage multi-scale, multi-level information comprehensively. The concatenated feature map is then fed into subsequent convolutional blocks to generate the final output feature map, which provides a rich feature representation for subsequent detection and classification tasks.

By integrating the CIB module, the model not only significantly decreases the number of parameters and computational costs but also accelerates inference speed. Importantly, it maintains robust feature extraction capabilities, thereby enhancing both the speed and efficiency of the model.

2.2. Partial Self-Attention

In this work, an efficient Partial Self-Attention (PSA) module [26] is introduced into the final layer of the backbone network, thereby improving feature extraction capabilities. Specifically, the proposed method first applies a 1 × 1 convolution to uniformly partition the features into two segments. Subsequently, only one segment is processed through the NPSA block, which is composed of a Multi-Head Self-Attention (MHSA) module and a Feed-Forward Network (FFN). The MHSA module captures diverse feature representations through multiple heads, thereby enhancing the model's expressive power. Following the approach in [26], the dimensions of the query and key in the MHSA are allocated to be half the size of the value dimension. The FFN then performs nonlinear transformations and feature reconstruction after the self-attention mechanism. Batch Normalization is employed instead of LayerNorm to accelerate inference speed and stabilize the training process. The two feature segments are subsequently concatenated and fused through a 1 × 1 convolution, ensuring the integrity and consistency of the features. This approach captures local detail while reducing computational complexity. The following outlines the specific implementation steps of the PSA module:

1. Input Feature Divisions:
   The input feature representation is evenly split into two segments using a 1 × 1 convolution. Assume that the input feature map X has the dimensions (B,C,H,W), where B represents the batch size, C the number of channels, and H and W the height and width, respectively. The PSA module first applies a 1 × 1 convolution to split the input feature map into two parts, each with $C/2$ channels, denoted as X_1 and X_2.

$$X_1, X_2 = \text{split}(\text{Convlxl}(X), \text{dim=1}) \tag{1}$$

2. Part Of The Input To The NPSA Module:
   One of the two divided feature parts, X_1 and X_2, is input into the NPSA module. The NPSA module consists of a Multi-Head Self-Attention (MHSA) mechanism and a Feed-Forward Network (FFN). Multi-Head Self-Attention (MHSA) is an extension of the self-attention mechanism, utilizing multiple attention heads to capture diverse feature representations. By allowing each head to focus on different aspects of the input data, MHSA enhances the model's capacity to extract a wider range of contextual information, thereby improving its ability to model complex dependencies and patterns within the data. This multi-headed approach provides richer and more robust feature learning compared to traditional single-head self-attention.
   Compute the Query, Key, and Value, and perform separate calculations using the multi-head mechanism:

$$Q = X_1 W_Q, K = X_1 W_K, V = X_1 W_V \tag{2}$$

   W_Q, W_K, and W_V are the weight matrices for the linear transformations.
   Compute the attention scores and perform weighted summation.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3}$$

where d_k is the dimension of the keys.
Multi-head mechanism: Concatenate the outputs of multiple heads and apply a linear transformation to restore the original dimensions.

$$\mathrm{MHSA}(X_1)=\mathrm{Linear}(\mathrm{Concat}(\mathrm{head}_1,\mathrm{head}_2,\ldots,\mathrm{head}_h)) \quad (4)$$

The Feed-Forward Network (FFN) generally comprises two fully connected layers along with an activation function (such as ReLU).

$$\mathrm{FFN}(x) = \mathrm{Linear}_2(\mathrm{ReLU}(\mathrm{Linear}_1(x))) \quad (5)$$

To accelerate inference speed, replace LayerNorm in MHSA with BatchNorm.

$$\mathrm{BatchNorm}(x) = \frac{x-\mu}{\sqrt{\sigma^2+\epsilon}} \quad (6)$$

3. Feature concatenation and fusion. Concatenate the features X_1' processed by the NPSA module with the unprocessed features X_2, and perform fusion using a 1×1 convolution.

$$Y = Convlx1(Concat(X_1', X_2)) \quad (7)$$

The PSA module is typically embedded within specific layers of the backbone network, such as following particular convolutional layers, to enhance the feature extraction process and improve the representational capacity of those layers. In this study, PSA is placed only after the lowest resolution Stage 4 to avoid excessive overhead from secondary complexities of self-attention. This approach introduces global representation learning capabilities into the YOLO model without significantly increasing computational costs, thereby enhancing the model's ability and performance. It achieves improved feature representation while maintaining computational efficiency, thus enhancing object detection performance.

2.3. Gold-YOLO

The backbone network is responsible for feature extraction, while the neck structure handles the fusion of these extracted feature vectors. However, information fusion issues still exist in the YOLOv8 model. In this study, we introduce an advanced collection-distribution mechanism from the Gold-YOLO [27] model to enhance the efficiency of information fusion. This mechanism leverages convolution and self-attention operations to strengthen multi-scale feature fusion capabilities, thereby improving the accuracy of object detection. Specifically, we adopt a lower-order collection-distribution mechanism to address the challenges of detecting small underwater objects. The Gold-YOLO network structure diagram is shown in Figure 3.

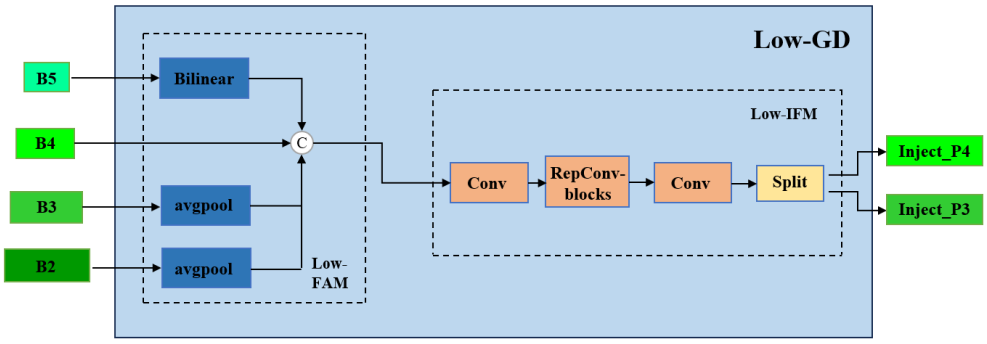


Figure 3. Structure diagram of the low-order collection-distribution mechanism.

The low-level collection-distribution mechanism integrates features from the B2, B3, B4, and B5 layers of the backbone, preserving high-resolution features crucial for small

object detection. Collection Process: The Low-FAM (Low-level Feature Alignment Module) employs average pooling to downsample the input features, aligning them to a uniform size. This step aligns and collects multi-scale features extracted by the backbone, ensuring efficient aggregation before passing them to the information fusion module. The Low-IFM (Low-level Information Fusion Module) then merges these aligned features to generate comprehensive global information. Distribution Process: The global information is subsequently distributed back to the lower-level feature maps through a simple attention mechanism, enhancing their representation capabilities. This approach allows the model to more effectively capture and recognize details in small objects, improving detection accuracy for such targets. By utilizing the low-level collection-distribution mechanism to gather and fuse information from various hierarchical layers, and redistributing it across different levels, the model enhances the detailed representation of lower-level feature maps. This method not only mitigates the inherent loss of information associated with conventional FPN architectures but also boosts the neck's feature fusion capabilities without significantly increasing latency, resulting in improved small object detection performance.

3. Experiments

3.1. Datasets

In this section, we utilize the custom-built underwater fusion dataset, CoopKnowledge, to validate the effectiveness of our proposed method improvements. The CoopKnowledge dataset incorporates a limited number of images from the RUOD open-source dataset along with images captured in an experimental pool using an underwater stereo camera. The dataset comprises 1077 underwater images spanning five categories: ROV, Coop models, fish, starfish, and sea urchins, all annotated with VOC format labels. Figure 4 provides example images from this dataset. For this study, we divided the CoopKnowledge dataset into training, validation, and test sets in a 7:2:1 ratio. The dataset example is shown in Figure 4.

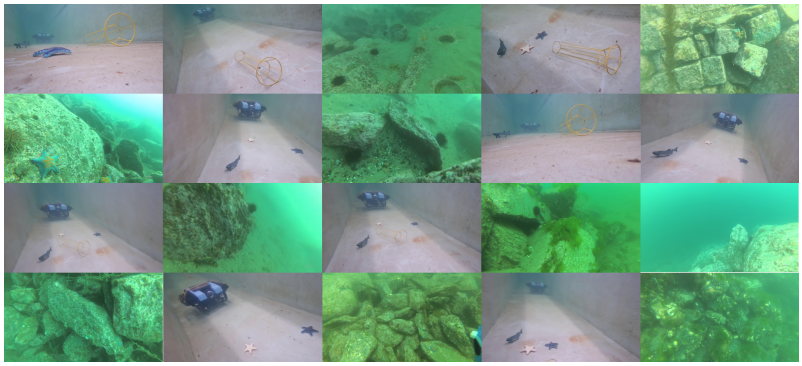


Figure 4. Some examples from CoopKnowledge's A dataset.

3.2. Evaluation Criteria

Precision and recall are indispensable metrics for evaluating model performance. Precision refers to the proportion of correctly predicted positive samples out of all samples predicted as positive by the model. In other words, it measures the accuracy of the model's positive predictions and can be interpreted as the false positive rate. Recall, on the other hand, refers to the ratio of actual positive samples that the model successfully identifies, indicating how many true positives were correctly identified, which can be interpreted as the false negative rate. The formulas for precision and recall are described as outlined below:

$$P = \frac{TP}{TP + FP}, R = \frac{TP}{TP + FN} \tag{8}$$

TP represents the number of true positive samples correctly predicted by the model, FP indicates the number of false positives where the model incorrectly predicts a positive

sample, and FN denotes the number of positive samples that were missed or not detected by the model.

The F1-Score takes both precision and recall into account, aiming to find a balance between these two metrics, ensuring that both are maximized simultaneously. It provides a comprehensive evaluation of the model's precision and recall performance. The formula for calculating the F1-Score is shown in Equation (9):

$$F_1 = \frac{2 \times P \times R}{P + R} \tag{9}$$

mAP50 refers to the mean Average Precision (mAP) calculated at a set Intersection over Union (IoU) threshold of 0.50, while mAP50-95 is the average precision computed over multiple IoU thresholds ranging from 0.50 to 0.95, typically in increments of 0.05. IoU represents the ratio of the area of overlap between the ground truth bounding box and the predicted bounding box to their total combined area. To calculate mAP, first, for each predicted bounding box, the IoU is calculated between the predicted and ground truth boxes. Next, a Precision-Recall (PR) curve is plotted based on the IoU values. Precision denotes the ratio of true positives to the total number of positive predictions, whereas recall signifies the ratio of true positives to the total number of actual positive instances. The area under the PR curve for each class is computed, which represents the Average Precision (AP). AP measures the average accuracy of the model's predictions for that particular class. Finally, the mAP is calculated by averaging the AP values across all classes, providing an overall assessment of the model's effectiveness in predicting objects throughout all categories. To mathematically express AP and mAP, let AP represent the average precision, *P(i)* the precision at recall level i, and *dR(i)* the change in recall between consecutive levels. The formula for AP and mAP is given as follows:

$$\mathrm{AP} = \sum_{i=1}^{n} P(i) dR(i) \tag{10}$$

$$\mathrm{mAP} = \frac{1}{n} \sum_{j=1}^{n} AP_j \tag{11}$$

3.3. Training Details

To validate the effects of the various improvements made to the YOLO model, we performed an ablation study on the suggested modifications. First, in order to enhance detection accuracy, we evaluated the feasibility of various modules and their combinations on the YOLOv8s baseline. These modules included the Compact Inverted Block (CIB), Partial Self-Attention Mechanism (PSA), and Gold-YOLO. Next, we compared our optimized model (YOLOv8s-CPG) with other models, such as Faster R-CNN and the YOLO series, which were trained under the same conditions. To ensure a fair evaluation, all models were trained from scratch without using pre-trained weights and trained for the same number of epochs using default settings.

This experiment was carried out on a laptop running the Windows 11 operating system, powered by a 13th Generation Intel® Core™ .We implemented the proposed network employing Python 3.9.7 and the PyTorch deep learning framework. Training continued until the model converged on the validation set, stopping after 150 epochs.

3.4. Results and Analysis

3.4.1. Ablation Experiments

To verify the effectiveness of the proposed improvements, we conducted multiple rigorous ablation experiments on our self-constructed underwater fusion dataset, CoopKnowledge. We trained the YOLOv8s-CPG model, the baseline model, and several other renowned detection networks on the CoopKnowledge dataset and evaluated their perfor-

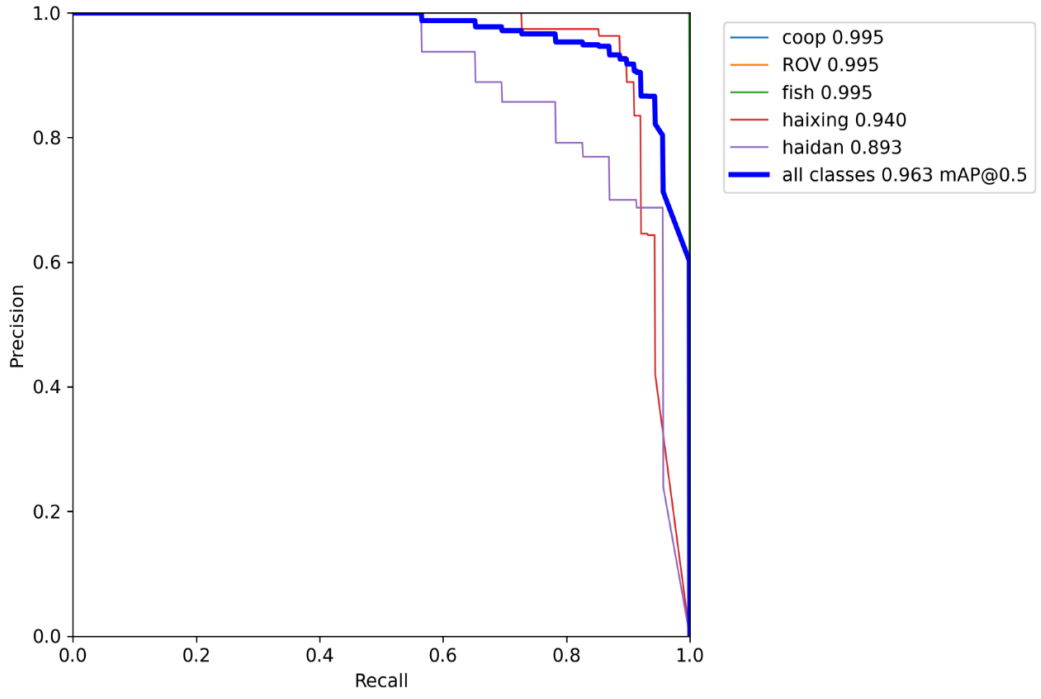
mance on the test set to ensure experimental fairness. The relevant experimental results are presented in Table 1, Figures 5 and 6.

Table 1. Ablation Experimental Data.

Models	Precision	Recall	mAP50	mAP50-90	F1
YOLOv8s	0.941	0.923	0.953	0.816	0.932
YOLOv8s-PSA	0.956	0.922	0.958	0.837	0.939
YOLOv8s-GY	0.943	0.927	0.956	0.834	0.935
YOLOv8s-CIB	0.959	0.931	0.959	0.841	0.945
YOLOv8s-CIB+PSA	0.960	0.933	0.958	0.838	0.940
YOLOv8s-CIB+GY	0.963	0.937	0.960	0.838	0.950
YOLOv8s-PSA+GY	0.969	0.936	0.961	0.842	0.953
YOLOv8s-CIB+PSA+GY	0.967	0.942	0.963	0.845	0.955

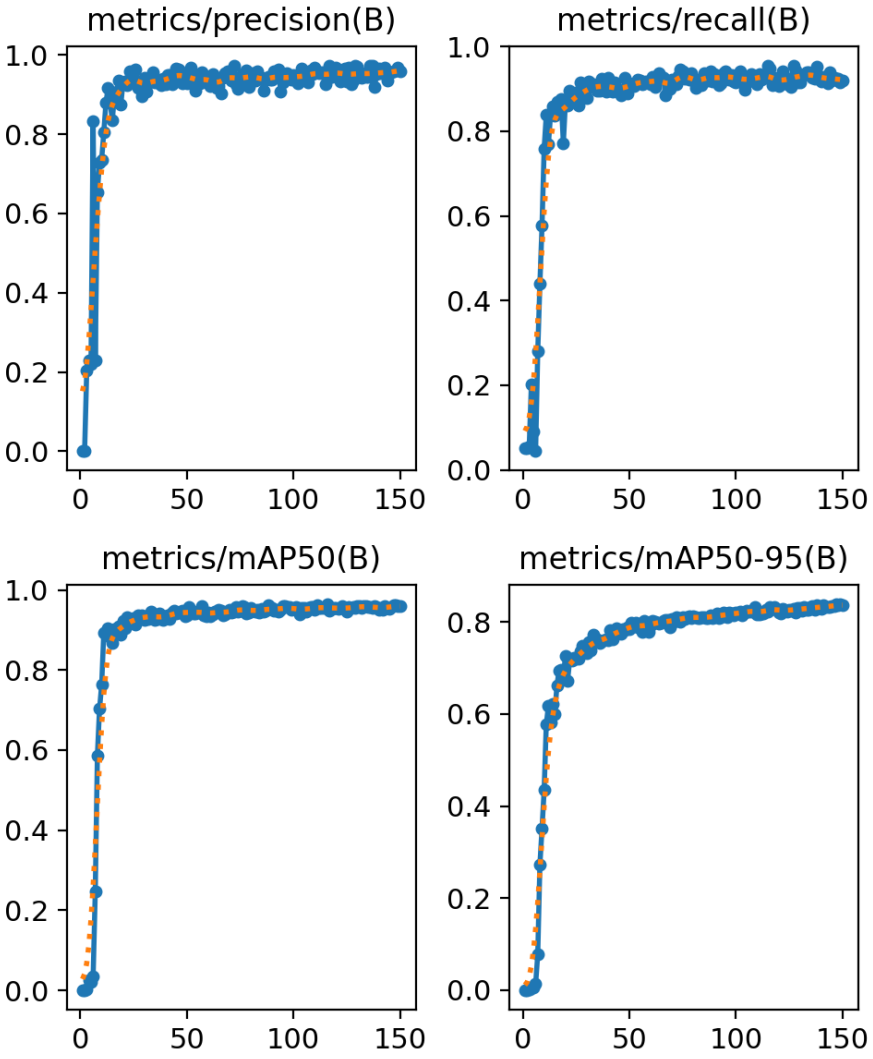
First, it is evident that the majority of the metrics from our improved model outperformed those of other ablation methods on the self-constructed underwater fusion dataset, particularly in terms of Recall, mAP50, and mAP50-95. In comparison to the original model, the enhanced version showed a marked increase in Precision, Recall, mAP50, and mAP50-95 by 2.76%, 2.06%, 1.05%, and 3.55%, respectively, demonstrating the advantage of the refined model in improving detection accuracy.

Next, when comparing the enhancement effects of single-module and dual-module approaches, it is evident that YOLOv8-CIB achieved optimal performance in underwater object recognition tasks. This is attributed to the use of large convolution kernels in the deep stages of the CIB module, which significantly improves the efficiency and accuracy of feature extraction. Among the dual-module models, YOLOv8-PSA+GY performed the best. The PSA enhanced the representational capacity of feature extraction by improving the display of details, while Gold-YOLO strengthened multi-scale feature fusion, allowing the extracted features to undergo more comprehensive multi-level integration. A comprehensive comparison of the data in the table indicates that CIB, PSA, and Gold-YOLO all contributed significantly to enhancing the prediction accuracy of the model.



(**a**)

Figure 5. *Cont.*



(**b**)

Figure 5. Experimental accuracy data results of YOLOv8s-CPG. (**a**) Precision data; (**b**) Detection accuracy data.

Additionally, by observing the curves in Figure 6, we noted that the improved model exhibited superior stability in mAP50 performance. The growth trend and convergence speed of this curve were faster than those of other improved submodules. During the initial growth phase of the curve, YOLOv8-PSA+CIB experienced significant fluctuations, mainly due to the relatively small receptive field of the convolutional kernel in the shallow structure of CIB during the early stages. Furthermore, in mAP50, we observed that the convergence speed of single-module PSA and single-module Gold-YOLO was similar to that of YOLOv8s. However, with the inclusion of CIB in YOLOv8s-CIB, the overall improvement was significantly enhanced. These experimental results underscore the effectiveness of the Compact Inverted Block (CIB).

Finally, based on the aforementioned advantages, the YOLOv8-CPG model integrates the feature representation capabilities of CIB and PSA, along with the multi-scale feature fusion functionality of Gold-YOLO. This integration significantly improves detection accuracy, making it more suitable for underwater target detection tasks.

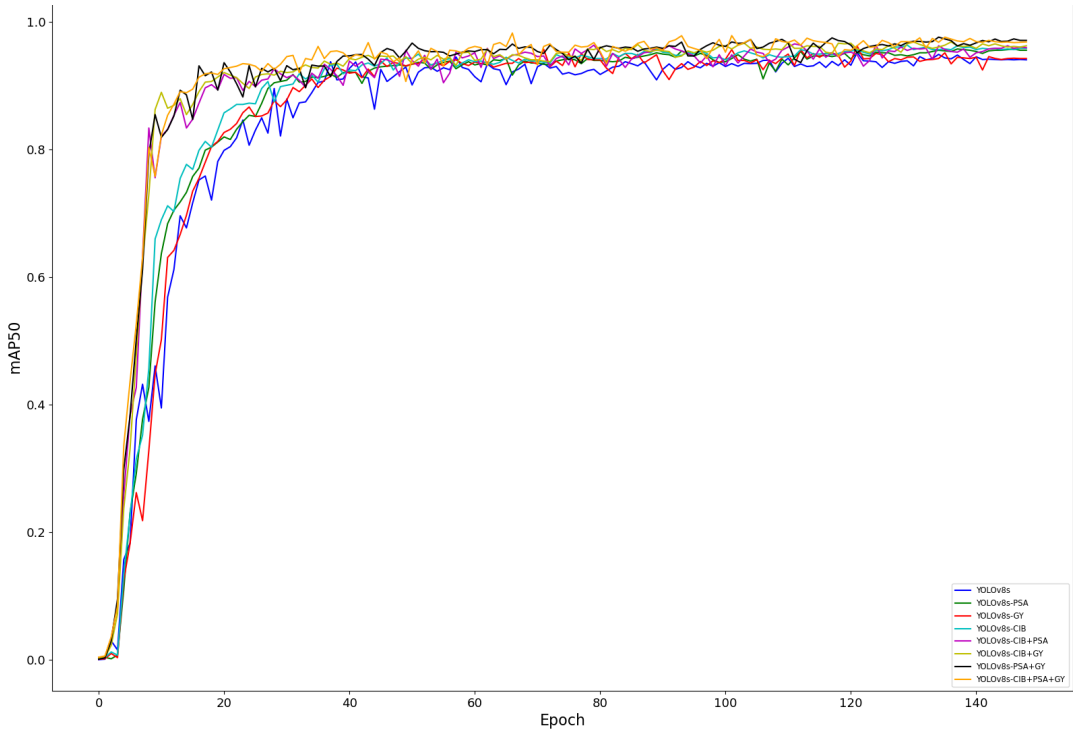


Figure 6. Ablation experimental data results of YOLOv8s-CIB+PSA+GY (YOLOv8s-CPG).

3.4.2. Comparative Experiments on Typical YOLO Models

In this section, we conduct a comparative study to evaluate the performance of six models (YOLOv3, YOLOv5, YOLOv6, YOLOv7, YOLOv11 and YOLOv8-CPG) over 150 training epochs for object detection tasks. The study focuses on key metrics, including Precision (P), Recall (R), mAP50, mAP50-95, and the F1 score.

First, upon analyzing the experimental data in Table 2, it is evident that the improved YOLOv8-CPG model outperforms the other models across all performance metrics. The outcomes of the experiments clearly indicate the superior capability of the proposed algorithm in underwater object detection, highlighting its robustness in this specific domain.

Table 2. Experimental Data Comparing Multiple Models.

Models	Precision	Recall	mAP50	mAP50-90	F1
YOLOv3s	0.917	0.909	0.897	0.792	0.913
YOLOv5s	0.936	0.913	0.941	0.807	0.925
YOLOv6s	0.937	0.922	0.947	0.812	0.930
YOLOv7s	0.944	0.918	0.952	0.821	0.931
YOLOv11s	0.963	0.937	0.960	0.839	0.950
YOLOv8s-CPG	0.967	0.942	0.963	0.845	0.955

Second, by examining the mAP50 data shown in Figure 7, we observe that the YOLOv8-CPG model demonstrates a more pronounced upward trend within the first 10 epochs. This suggests that the method not only converges faster but also maintains better stability throughout the training process. YOLOv11 demonstrates a fast convergence rate in the initial stages, achieving gradual stability between the 10th and 30th iterations. However, its performance, in terms of both curve values and precision, is slightly lower compared to the YOLOv8-CPG model. While YOLOv6 initially exhibits some fluctuations in precision and recall, its overall performance curve remains higher than YOLOv5. However, between epochs 10 and 40, YOLOv5 shows a more rapid convergence rate.

The results reveal that the improved method offers both enhanced training speed and greater stability. This advantage can be attributed to the proposed algorithm's ability to better capture the target's features and optimize the model architecture. In particular, the integration of the Compact Inverted Block (CIB), PSA self-attention mechanism, and Gold-YOLO-based multi-scale feature fusion plays a pivotal role in boosting the model's training effectiveness.

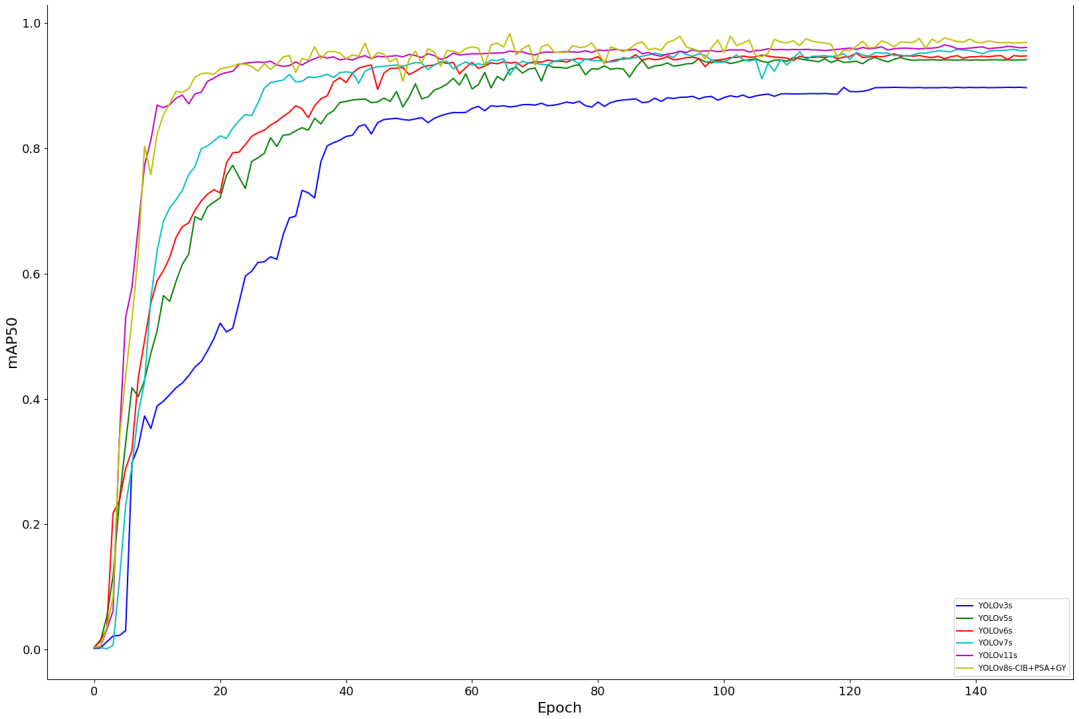


Figure 7. Comparison of model indicators of yolo series.

In conclusion, the proposed enhancements demonstrate superior accuracy and stability for object recognition tasks in complex underwater environments. These results not only underscore the theoretical significance of the improvements but also highlight their practical value in real-world applications. The refinements introduced in this paper provide robust support for advancing model performance in challenging detection tasks.

3.4.3. Experimental Results of the Improved Algorithm on the Test Set

To validate the detection capability of the improved model in real-world scenarios we conducted experiments using the preserved real-world test set from our dataset, along with the WildFish dataset, to evaluate the applicability of the improved algorithm in dynamic water conditions and uncontrolled underwater environments. These experiments were designed to assess the algorithm's performance under realistic scenarios. The comparison between the baseline model and the improved algorithm is shown in the figure below. Through comparative experiments, we evaluated the performance of five models (YOLOv8, YOLOv9, YOLOv10, YOLOv11, and YOLOv8-CPG) over 150 training epochs for object detection. The comparison focused on metrics such as Precision (P), Recall (R), F1 score, mAP50, and mAP50-95. The results, presented in Table 3, indicate that the YOLOv8-CPG model slightly outperforms the other YOLO series models in terms of overall detection accuracy. The results clearly demonstrate the exceptional performance of YOLOv8-CPG in detection tasks. The model achieved an average detection accuracy of 0.839, representing a 7.3% improvement over the baseline YOLOv8 model, which had an average accuracy of 0.782.

As observed in Figure 8, YOLOv8-CPG consistently outperformed the baseline model, even in demanding underwater conditions marked by low lighting and blurred distinctions between the target and the background. As shown in Figure 9, YOLO8-CPG maintains a commendable level of precision on the WildFish dataset, which was captured in real underwater environments. Moreover, its accuracy surpasses that of the baseline model. This result further verifies the effectiveness of the suggested improvements in addressing underwater object detection tasks and highlights the model's capability to handle real-world scenarios in such environments.

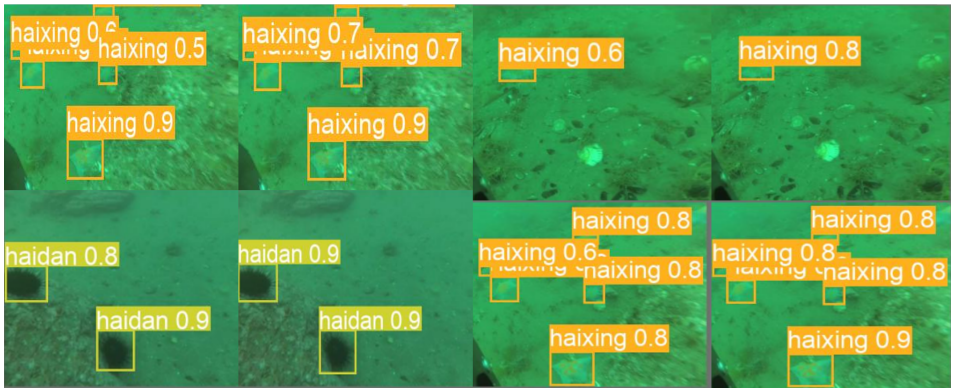


Figure 8. Comparison of YOLO8-CPG in a real underwater environment.

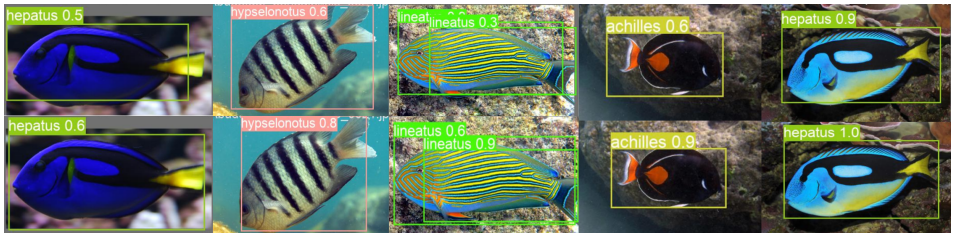


Figure 9. Comparison of YOLO8-CPG on the WildFish dataset.

Table 3. Comparison of the accuracy of YOLO8-CPG in the WildFish fish dataset.

Models	Precision	Recall	mAP50	mAP50-90	F1
YOLOv8s	0.782	0.769	0.762	0.675	0.776
YOLOv9s	0.808	0.793	0.805	0.719	0.801
YOLOv10s	0.831	0.811	0.827	0.752	0.821
YOLOv11s	0.835	0.815	0.830	0.771	0.825
YOLOv8s-CPG	0.839	0.821	0.833	0.768	0.830

4. Conclusions

In this study, we propose an advanced deep learning model named YOLOv8-CPG, which is designed to tackle the challenge of low detection accuracy in underwater object detection. This model introduces the C2f module into the backbone architecture of YOLOv8 and utilizes larger convolutional kernels in deeper layers, which, with their greater receptive field, enable the more accurate identification of object contours. This leads to improvements in detection accuracy and efficiency. The model also maintains strong feature extraction capabilities while improving inference speed and efficiency. Additionally, the integration of the PSA attention mechanism and the use of the Gold-YOLO low-level collection-distribution mechanism enhance the neck structure of YOLOv8, allowing for better focus on underwater target features and improving multi-scale feature fusion capacity, ultimately boosting detection accuracy and overall model performance.

The experimental results indicate that YOLOv8-CPG offers performance enhancements across key metrics, including Precision, Recall, mAP50, and mAP50-95, compared to the traditional YOLOv8 model. Specifically, Precision and Recall saw increases of 2.76%

and 2.06%, respectively, while mAP50 and mAP50-95 rose by 1.05% and 3.55%. These improvements demonstrate that the method not only enhances detection accuracy but also maintains stability in performance, providing critical technical support for accurate object recognition in underwater environments.

In conclusion, the proposed YOLOv8-CPG method achieves significant improvements in underwater object detection, offering a more reliable and efficient solution for future autonomous underwater vehicle (AUV) applications. Moving forward, we plan to further optimize this approach and explore its applicability to other underwater scenarios, broadening its range of use cases to meet diverse operational requirements.

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Abbreviations

The abbreviations listed below are utilized in this manuscript:

C2f	CSPDarknet53 to 2-Stage FPN
CIB	Compact Inverted Block
PSA	Partial Self-Attention
YOLOv8-CPG	YOLOv8-CIB+PSA+Gold-YOLO
MHSA	Multi-Head Self-Attention
FFN	Feed-Forward Network
Low-FAM	Low-level Feature Alignment Module

References

1. Li, Z.; Cai, D.; Wang, J.; Li, Y.; Gui, G.; Sun, X.; Wang, N.; Zhang, J.; Liu, H.; Wang, G. Machine learning based dynamic correlation on marine environmental data using cross-recurrence strategy. *IEEE Access* **2019**, *7*, 185121–185130. [CrossRef]
2. Qin, X.; Luo, X.; Wu, Z.; Shang, J.; Zhao, D. Deep learning-based high accuracy bottom tracking on 1-D side-scan sonar data. *IEEE Geosci. Remote Sens. Lett.* **2021**, *19*, 8011005. [CrossRef]
3. Girshick, R.; Donahue, J.; Darrell, T.; Malik, J. Rich feature hierarchies for accurate object detection and semantic segmentation. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Columbus, OH, USA, 23 June 2014; pp. 580–587.
4. Girshick, R. Fast r-cnn. *arXiv* **2015**, arXiv:1504.08083.
5. Ren, S. Faster r-cnn: Towards real-time object detection with region proposal networks. *arXiv* **2015**, arXiv:1506.01497. [CrossRef] [PubMed]
6. Liu, W.; Anguelov, D.; Erhan, D.; Szegedy, C.; Reed, S.; Fu, C.-Y.; Berg, A.C. Ssd: Single shot multibox detector. In Proceedings of the Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, 11 October 2016; pp. 21–37.
7. Jiang, P.; Ergu, D.; Liu, F.; Cai, Y.; Ma, B. A Review of Yolo algorithm developments. *Procedia Comput. Sci.* **2022**, *199*, 1066–1073. [CrossRef]
8. Hsiao, Y.-H.; Chen, C.-C.; Lin, S.-I.; Lin, F.-P. Real-world underwater fish recognition and identification, using sparse representation. *Ecol. Inform.* **2014**, *23*, 13–21. [CrossRef]

9. Li, X.; Shang, M.; Qin, H.; Chen, L. Fast accurate fish detection and recognition of underwater images with fast r-cnn. In Proceedings of the OCEANS 2015-MTS/IEEE Washington, Washington, DC, USA, 22 October 2015; pp. 1–5.
10. Salman, A.; Jalal, A.; Shafait, F.; Mian, A.; Shortis, M.; Seager, J.; Harvey, E. Fish species classification in unconstrained underwater environments based on deep learning. *Limnol. Oceanogr. Methods* **2016**, *14*, 570–585. [CrossRef]
11. Zhuang, P.; Xing, L.; Liu, Y.; Guo, S.; Qiao, Y. Marine Animal Detection and Recognition with Advanced Deep Learning Models. In Proceedings of the CLEF (Working Notes), Dublin, Ireland, 11 September 2017; pp. 166–177.
12. Zhao, J.; Bao, W.; Zhang, F.; Zhu, S.; Liu, Y.; Lu, H.;Shen, M.; Ye, Z. Modified motion influence map and recurrent neural network-based monitoring of the local unusual behaviors for fish school in intensive aquaculture. *Aquaculture* **2018**, *493*, 165–175. [CrossRef]
13. Zhou, C.; Xu, D.; Chen, L.; Zhang, S.; Sun, C.; Yang, X.; Wang, Y. Evaluation of fish feeding intensity in aquaculture using a convolutional neural network and machine vision. *Aquaculture* **2019**, *507*, 457–465. [CrossRef]
14. Adiwinata, Y.; Sasaoka, A.; Bayupati, I.A.; Sudana, O. Fish species recognition with faster r-cnn inception-v2 using qut fish dataset. *Lontar Komput. J. Ilm. Teknol. Inf.* **2020**, *11*, 144–154. [CrossRef]
15. Zhao, S.; You, F. Vehicle detection based on improved yolov3 algorithm. In Proceedings of the 2020 International Conference on Intelligent Transportation, Big Data & Smart City (ICITBS), Vientiane, Laos, 11–14 January 2020; pp. 76–79.
16. Lu, Z.;Ding, L.; Wang, Z.; Dong, L.; Guo, Z. Road condition detection based on deep learning YOLOv5 network. In Proceedings of the 2023 IEEE 3rd International Conference on Electronic Technology, Communication and Information (ICETCI), Changchun, China, 26–28 May 2023; pp. 497–501.
17. Wang, Y.; Pan, F.; Li, Z.; Xin, X.; Li, W. CoT-YOLOv8: Improved YOLOv8 for Aerial images Small Target Detection. In Proceedings of the 2023 China Automation Congress (CAC), Chongqing, China, 17–19 November 2023; pp. 4943–4948.
18. Cai, K.; Miao, X.; Wang, W.; Pang, H.; Liu, Y.; Song, J. A modified YOLOv3 model for fish detection based on MobileNetv1 as backbone. *Aquac. Eng.* **2020**, *91*, 102117. [CrossRef]
19. Yu, G.; Cai, R.; Su, J.; Hou, M.; Deng, R. U-YOLOv7: A network for underwater organism detection. *Ecol. Inform.* **2023**, *75*, 102108. [CrossRef]
20. Cai, Y.; Yao, Z.; Jiang, H.; Qin, W.; Xiao, J.; Huang, X.; Pan, J.; Feng, H. Rapid detection of fish with SVC symptoms based on machine vision combined with a NAM-YOLO v7 hybrid model. *Aquaculture* **2024**, *582*, 740558. [CrossRef]
21. Yi, W.; Wang, B. Research on Underwater small target Detection Algorithm based on improved YOLOv7. *IEEE Access* **2023**, *11*, 66818–66827. [CrossRef]
22. Qu, S.; Cui, C.; Duan, J.; Lu, Y.; Pang, Z. Underwater small target detection under YOLOv8-LA model. *Sci. Rep.* **2024**, *14*, 16108. [CrossRef] [PubMed]
23. Zhu, Y.; Yin, H.; Hu, M. An Improved Underwater Target Detection Algorithm Based on YOLOv8. In Proceedings of the 2023 2nd International Conference on Artificial Intelligence, Human-Computer Interaction and Robotics (AIHCIR), Tianjin, China, 8–10 December 2023; pp. 521–525.
24. Ultralytics YOLOv8. Available online: https://docs.ultralytics.com/#ultralytics-yolov8 (accessed on 10 May 2023).
25. Wang, A.; Chen, H.; Liu, L.; Chen, K.; Lin, Z.; Han, J.; Ding, G. Yolov10: Real-time end-to-end object detection. *arXiv* **2024**, arXiv:2405.14458.
26. Graham, B.; El-Nouby, A.; Touvron, H.; Stock, P.; Joulin, A.; Jégou, H.; Douze, M. Levit: A vision transformer in convnet's clothing for faster inference. In Proceedings of the IEEE/CVF International Conference on Computer Vision, Montreal, QC, Canada, 10 March 2021; pp. 12259–12269.
27. Wang, C.; He, W.; Nie, Y.; Guo, J.; Liu, C.; Wang, Y.; Han, K. Gold-YOLO: Efficient object detector via gather-and-distribute mechanism. *Adv. Neural Inf. Process. Syst.* **2024**, *36*.

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